

Azure Synapse SQL Pool to Microsoft Fabric OneLake Migration Lab

Objective

In this lab, you will:

- Provision an Azure Synapse Analytics workspace with a dedicated SQL pool
- Populate the dedicated SQL pool with sample sales data
- Connect Microsoft Fabric to the Azure Synapse dedicated SQL pool
- Ingest data into a Fabric Lakehouse backed by OneLake
- Validate the migrated data through the Lakehouse SQL analytics endpoint
- Perform basic data transformation using PySpark
- Save and verify the transformed data as a Delta table in OneLake

Phase 1: Provision Azure Synapse Analytics

Step 1: Create a Synapse Analytics Workspace

1. Navigate to **Azure Synapse Analytics**.
2. Click **Create**.
3. Configure the following:

Setting	Value
Workspace Name	synapsefabricws001 (or any unique name)
Region	Same region as your Fabric capacity
ADLS Gen2 Account	synapsedatalake001
File System Name	synapsefs
SQL Admin Login	sqladminuser
Password	Your preferred secure password

1. Click **Review + Create**.
2. Deploy the workspace.

Step 2: Create a Dedicated SQL Pool

1. Open **Synapse Studio** from the workspace overview.
2. Go to **Manage** → **SQL pools** → **New**.
3. Configure the pool:

Setting	Value
Dedicated SQL Pool Name	sqlpool_sales
Performance Level	DW100c (lowest tier for the lab)

1. Review the configuration.
2. Click **Create** and wait for the pool to reach the **Online** state.

Note: A dedicated SQL pool (formerly SQL DW) is a Massively Parallel Processing (MPP) engine. Remember to pause the pool after the lab to avoid unnecessary charges.

Phase 2: Create Sample Data in the Dedicated SQL Pool

Step 3: Open a SQL Script

1. In Synapse Studio, go to **Develop** → + → **SQL script**.
2. In **Connect to**, select **sqlpool_sales**.

Step 4: Create the SalesOrders Table

Execute the following script. The table is created as a round-robin HEAP, which is suitable for a small staging table:

```
CREATE TABLE dbo.SalesOrders
(
    OrderID          INT          NOT NULL,
    CustomerName     VARCHAR(100) NOT NULL,
    Region           VARCHAR(50)  NOT NULL,
    Product          VARCHAR(50)  NOT NULL,
    Amount           INT          NOT NULL,
    OrderDate        DATE         NOT NULL
)
WITH
(
    DISTRIBUTION = ROUND_ROBIN,
    HEAP
);
```

Step 5: Insert Sample Records

Run the following statement to load 24 rows:

```
INSERT INTO dbo.SalesOrders VALUES
(1, 'Alex',      'North', 'Laptop',  85000, '2026-01-05'),
(2, 'Amit',      'South', 'Monitor', 18000, '2026-01-08'),
(3, 'John',      'East',   'Server', 240000, '2026-01-11'),
(4, 'Sarah',     'West',   'Laptop', 92000, '2026-01-15'),
(5, 'Priya',     'North', 'Keyboard', 3500, '2026-01-18'),
(6, 'Michael',   'South', 'Server', 210000, '2026-01-22'),
(7, 'Neha',      'East',   'Laptop', 78000, '2026-01-26'),
(8, 'David',     'West',   'Monitor', 16000, '2026-02-02'),
(9, 'Anjali',    'North', 'Tablet', 42000, '2026-02-06'),
(10, 'Robert',   'South', 'Laptop', 88000, '2026-02-09'),
(11, 'Sneha',    'East',   'Keyboard', 4200, '2026-02-13'),
(12, 'Chris',    'West',   'Server', 265000, '2026-02-17'),
(13, 'Rohan',    'North', 'Laptop', 95000, '2026-02-21'),
(14, 'Emma',     'South', 'Tablet', 38000, '2026-02-25'),
(15, 'Vikram',   'East',   'Monitor', 21000, '2026-03-01'),
(16, 'Olivia',   'West',   'Laptop', 81000, '2026-03-05'),
(17, 'Arjun',    'North', 'Server', 198000, '2026-03-09'),
(18, 'Sophia',   'South', 'Keyboard', 3900, '2026-03-13'),
(19, 'Karan',    'East',   'Tablet', 45000, '2026-03-17'),
(20, 'Isabella', 'West',   'Laptop', 99000, '2026-03-21'),
```

```
(21, 'Rahul', 'North', 'Monitor', 17500, '2026-03-25'),
(22, 'Mia', 'South', 'Server', 232000, '2026-03-29'),
(23, 'Daniel', 'East', 'Laptop', 76000, '2026-04-02'),
(24, 'Pooja', 'West', 'Tablet', 41000, '2026-04-06');
```

Step 6: Validate the Data

Execute:

```
SELECT * FROM dbo.SalesOrders;
```

Confirm that 24 rows are returned before proceeding.

Phase 3: Connect Microsoft Fabric to Azure Synapse

Step 7: Create a New Connection

1. Open your **Fabric Workspace**.
2. Navigate to **Manage Connections and Gateways** → **New Connection**.
3. Select **Azure SQL Database** (the dedicated SQL pool is reached through its SQL endpoint).
4. Provide the following information:

Parameter	Value
Server Name	synapsefabricws001.sql.azuresynapse.net
Database Name	sqlpool_sales
Authentication	Basic (SQL Authentication)
Username	sqladminuser
Password	Your SQL Password

1. Click **Test Connection**.
2. Verify the connection is successful.
3. Click **Create**.

Tip: The dedicated SQL pool endpoint ends in .sql.azuresynapse.net (not .database.windows.net). Ensure the firewall rule “Allow Azure services and resources to access this workspace” is enabled.

Phase 4: Ingest Data into Microsoft Fabric OneLake

Step 8: Create a Lakehouse

1. In the Fabric Workspace, click **New Item**.
2. Select **Lakehouse**.
3. Enter:

Setting	Value
Lakehouse Name	LH_SalesData

1. Create the Lakehouse.

About OneLake: Every Lakehouse is automatically backed by OneLake, the single, tenant-wide data lake in Fabric. Tables you create land as Delta-Parquet files under the Lakehouse Tables/ folder in OneLake — no separate storage account to provision.

Step 9: Create a Data Pipeline

1. Click **New Item**.
2. Select **Data Pipeline**.
3. Configure:

Setting	Value
Pipeline Name	PL_SynapseToOneLake

1. Create the pipeline.

Step 10: Configure the Copy Data Activity

Add a **Copy data** activity to the pipeline, then configure source and destination.

Source Configuration

Setting	Value
Connection	synapsefabricws001 (Azure SQL)
Source Type	Azure SQL / Synapse dedicated SQL pool
Use Query	Table
Table	dbo.SalesOrders

Destination Configuration

Setting	Value
Destination Type	Fabric Lakehouse (OneLake)
Lakehouse	LH_SalesData
Root Folder	Tables
Target Table	SalesOrders
Table Action	Overwrite

Save the pipeline configuration.

Step 11: Execute the Pipeline

1. Click **Run**.
2. Monitor execution progress.
3. Wait until the pipeline status displays **Succeeded**.

Phase 5: Validate Data in Fabric OneLake

Step 12: Verify Data in the Lakehouse SQL Endpoint

1. Open the **LH_SalesData** Lakehouse.
2. Switch to the **SQL analytics endpoint**.
3. Execute:

```
SELECT * FROM SalesOrders;
```

Expected Result

The query returns the same 24 records that exist in the Azure Synapse dedicated SQL pool. You can also confirm the physical files by browsing the Lakehouse Tables/SalesOrders folder in OneLake.

Phase 6: Transform Data Using PySpark

Step 13: Read Data into a Fabric Notebook

1. Create a new **Notebook**.
2. Attach the notebook to the **LH_SalesData** Lakehouse.
3. Execute:

```
df = spark.read.table("SalesOrders")
display(df)
```

Verify that all 24 sales records are displayed.

Step 14: Add an OrderTier Column

Run the following PySpark code to classify orders by value:

```
from pyspark.sql.functions import when, col

df2 = df.withColumn(
    "OrderTier",
    when(col("Amount") >= 100000, "Enterprise")
    .when(col("Amount") >= 40000, "Mid-Market")
    .otherwise("SMB")
)

display(df2)
```

Transformation Logic

Amount	OrderTier
>= 100000	Enterprise
40000 - 99999	Mid-Market
< 40000	SMB

Step 15: Save the Transformed Data

Execute:

```
df2.write.mode("overwrite").saveAsTable("SalesOrders_Transformed")
```

This creates a new Delta table in OneLake named **SalesOrders_Transformed**.

Step 16: Validate the Transformed Data

Open the Lakehouse SQL analytics endpoint and execute:

```
SELECT * FROM SalesOrders_Transformed;
```

Expected Result

All 24 rows are returned with the additional OrderTier column populated, confirming a successful end-to-end migration and transformation from the Azure Synapse dedicated SQL pool into Fabric OneLake.